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### Methods and devices for video rendering for video see-through (VST) augmented reality (AR)

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#### Abstract

A method includes capturing an image and associating the image with a camera pose for each of multiple cameras. The method also includes determining, for each camera, a first contribution of the image for a first virtual view for display on a first display and a second contribution of the image for a second virtual view for display on a second display. The method further includes determining, for each camera, a first confidence map for the first virtual view based on the camera pose and a position of the camera in relation to a first virtual camera and a second confidence map for the second virtual view based on the camera pose and the position of the camera in relation to a second virtual camera. In addition, the method includes generating the first virtual view by combining the first contribution using the first confidence map for each of the cameras and the second virtual view by combining the second contribution using the second confidence map for each of the cameras.

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**USPC:** None

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION AND PRIORITY CLAIM (1) This application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 63/355,373 filed on Jun. 24, 2022, which is hereby incorporated by reference in its entirety.

### TECHNICAL FIELD

(1) This disclosure relates generally to augmented reality (AR) devices and processes. More specifically, this disclosure relates to methods and devices for video rendering for video see-through (VST) AR.

## BACKGROUND

(2) Augmented reality (AR) systems can seamlessly blend virtual objects generated by computer graphics within real-world scenes. Optical see-through (OST) AR systems refer to AR systems in which users directly view real-world scenes through head-mounted devices (HMDs).

Unfortunately, OST AR systems face many challenges that can limit their adoption. Some of these challenges include limited fields of view, limited usage spaces (such as indoor-only usage), failure to display fully-opaque black objects, and usage of complicated optical pipelines that may require projectors, waveguides, and other optical elements.

## SUMMARY

(3) This disclosure provides methods and devices for video rendering for video see-through (VST) augmented reality (AR).

(4) In a first embodiment, a method for video rendering for VST AR on an AR device includes capturing an image and associating the captured image with a camera pose for each of a plurality of cameras of the AR device. The method also includes determining, for each of the cameras, (i) a first contribution of the captured image for a first virtual view for display on a first display of the AR device and (ii) a second contribution of the captured image for a second virtual view for display on a second display of the AR device. The method further includes determining, for each of the cameras, (i) a first confidence map for the first virtual view based on the camera pose and a position of the camera in relation to a first virtual camera and (ii) a second confidence map for the second virtual view based on the camera pose and the position of the camera in relation to a second virtual camera. In addition, the method includes generating (i) the first virtual view by combining the first contribution using the first confidence map for each of the plurality of cameras and (ii) the second virtual view by combining the second contribution using the second confidence map for each of the plurality of cameras.

(5) In a second embodiment, a VST AR device includes a plurality of cameras and at least one processing device operably coupled to the cameras. The at least one processing device is configured, for each of the cameras, to capture an image using the camera and associate the captured image with a camera pose. The at least one processing device is also configured to determine, for each of the cameras, (i) a first contribution of the captured image for a first virtual view for display on a first display of the AR device and (ii) a second contribution of the captured image for a second virtual view for display on a second display of the AR device. The at least one processing device is further configured to determine, for each of the cameras, (i) a first confidence map for the first virtual view based on the camera pose and a position of the camera in relation to a first virtual camera and (ii) a second confidence map for the second virtual view based on the camera pose and the position of the camera in relation to a second virtual camera. In addition, the at least one processing device is configured to generate (i) the first virtual view by combining the first contribution using the first confidence map for each of the plurality of cameras and (ii) the second virtual view by combining the second contribution using the second confidence map for each of the plurality of cameras.

(6) In a third embodiment, a non-transitory machine readable medium contains instructions that when executed cause at least one processor to capture an image and associate the captured image with a camera pose for each of a plurality of cameras of a VST AR device. The non-transitory machine readable medium also contains instructions that when executed cause the at least one processor to determine, for each of the cameras, (i) a first contribution of the captured image for a first virtual view for display on a first display of the AR device and (ii) a second contribution of the captured image for a second virtual view for display on a second display of the AR device. The non-transitory machine readable medium further contains instructions that when executed cause the at least one processor to determine, for each of the cameras, (i) a first confidence map for the first virtual view based on the camera pose and a position of the camera in relation to a first virtual camera and (ii) a second confidence map for the second virtual view based on the camera pose and

the position of the camera in relation to a second virtual camera. In addition, the non-transitory machine readable medium contains instructions that when executed cause the at least one processor to generate (i) the first virtual view by combining the first contribution using the first confidence map for each of the plurality of cameras and (ii) the second virtual view by combining the second contribution using the second confidence map for each of the plurality of cameras.

(7) Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

(8) Before undertaking the DETAILED DESCRIPTION below, it may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The terms “transmit,” “receive,” and “communicate,” as well as derivatives thereof, encompass both direct and indirect communication. The terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation. The term “or” is inclusive, meaning and/or. The phrase “associated with,” as well as derivatives thereof, means to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like.

(9) Moreover, various functions described below can be implemented or supported by one or more computer programs, each of which is formed from computer readable program code and embodied in a computer readable medium. The terms “application” and “program” refer to one or more computer programs, software components, sets of instructions, procedures, functions, objects, classes, instances, related data, or a portion thereof adapted for implementation in a suitable computer readable program code. The phrase “computer readable program code” includes any type of computer code, including source code, object code, and executable code. The phrase “computer readable medium” includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. A “non-transitory” computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable memory device.

(10) As used here, terms and phrases such as “have,” “may have,” “include,” or “may include” a feature (like a number, function, operation, or component such as a part) indicate the existence of the feature and do not exclude the existence of other features. Also, as used here, the phrases “A or B,” “at least one of A and/or B,” or “one or more of A and/or B” may include all possible combinations of A and B. For example, “A or B,” “at least one of A and B,” and “at least one of A or B” may indicate all of (1) including at least one A, (2) including at least one B, or (3) including at least one A and at least one B. Further, as used here, the terms “first” and “second” may modify various components regardless of importance and do not limit the components. These terms are only used to distinguish one component from another. For example, a first user device and a second user device may indicate different user devices from each other, regardless of the order or importance of the devices. A first component may be denoted a second component and vice versa without departing from the scope of this disclosure.

(11) It will be understood that, when an element (such as a first element) is referred to as being (operatively or communicatively) “coupled with/to” or “connected with/to” another element (such as a second element), it can be coupled or connected with/to the other element directly or via a third element. In contrast, it will be understood that, when an element (such as a first element) is referred to as being “directly coupled with/to” or “directly connected with/to” another element (such as a second element), no other element (such as a third element) intervenes between the element and the other element.

(12) As used here, the phrase “configured (or set) to” may be interchangeably used with the phrases

“suitable for,” “having the capacity to,” “designed to,” “adapted to,” “made to,” or “capable of” depending on the circumstances. The phrase “configured (or set) to” does not essentially mean “specifically designed in hardware to.” Rather, the phrase “configured to” may mean that a device can perform an operation together with another device or parts. For example, the phrase “processor configured (or set) to perform A, B, and C” may mean a generic-purpose processor (such as a CPU or application processor) that may perform the operations by executing one or more software programs stored in a memory device or a dedicated processor (such as an embedded processor) for performing the operations.

(13) The terms and phrases as used here are provided merely to describe some embodiments of this disclosure but not to limit the scope of other embodiments of this disclosure. It is to be understood that the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. All terms and phrases, including technical and scientific terms and phrases, used here have the same meanings as commonly understood by one of ordinary skill in the art to which the embodiments of this disclosure belong. It will be further understood that terms and phrases, such as those defined in commonly-used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined here. In some cases, the terms and phrases defined here may be interpreted to exclude embodiments of this disclosure.

(14) Examples of an “electronic device” according to embodiments of this disclosure may include at least one of a smartphone, a tablet personal computer (PC), a mobile phone, a video phone, an e-book reader, a desktop PC, a laptop computer, a netbook computer, a workstation, a personal digital assistant (PDA), a portable multimedia player (PMP), an MP3 player, a mobile medical device, a camera, or a wearable device (such as smart glasses, a head-mounted device (HMD), electronic clothes, an electronic bracelet, an electronic necklace, an electronic accessory, an electronic tattoo, a smart mirror, or a smart watch). Other examples of an electronic device include a smart home appliance. Examples of the smart home appliance may include at least one of a television, a digital video disc (DVD) player, an audio player, a refrigerator, an air conditioner, a cleaner, an oven, a microwave oven, a washer, a drier, an air cleaner, a set-top box, a home automation control panel, a security control panel, a TV box (such as SAMSUNG HOMESYNC, APPLETV, or GOOGLE TV), a smart speaker or speaker with an integrated digital assistant (such as SAMSUNG GALAXY HOME, APPLE HOMEPOD, or AMAZON ECHO), a gaming console (such as an XBOX, PLAYSTATION, or NINTENDO), an electronic dictionary, an electronic key, a camcorder, or an electronic picture frame. Still other examples of an electronic device include at least one of various medical devices (such as diverse portable medical measuring devices (like a blood sugar measuring device, a heartbeat measuring device, or a body temperature measuring device), a magnetic resource angiography (MRA) device, a magnetic resource imaging (MRI) device, a computed tomography (CT) device, an imaging device, or an ultrasonic device), a navigation device, a global positioning system (GPS) receiver, an event data recorder (EDR), a flight data recorder (FDR), an automotive infotainment device, a sailing electronic device (such as a sailing navigation device or a gyro compass), avionics, security devices, vehicular head units, industrial or home robots, automatic teller machines (ATMs), point of sales (POS) devices, or Internet of Things (IoT) devices (such as a bulb, various sensors, electric or gas meter, sprinkler, fire alarm, thermostat, street light, toaster, fitness equipment, hot water tank, heater, or boiler). Other examples of an electronic device include at least one part of a piece of furniture or building/structure, an electronic board, an electronic signature receiving device, a projector, or various measurement devices (such as devices for measuring water, electricity, gas, or electromagnetic waves). Note that, according to various embodiments of this disclosure, an electronic device may be one or a combination of the above-listed devices. According to some embodiments of this disclosure, the electronic device may be a flexible electronic device. The electronic device disclosed here is not limited to the above-listed devices and may include any other electronic devices now known or later developed.

(15) In the following description, electronic devices are described with reference to the accompanying drawings, according to various embodiments of this disclosure. As used here, the term “user” may denote a human or another device (such as an artificial intelligent electronic device) using the electronic device.

(16) Definitions for other certain words and phrases may be provided throughout this patent document. Those of ordinary skill in the art should understand that in many if not most instances, such definitions apply to prior as well as future uses of such defined words and phrases.

(17) None of the descriptions in this application should be read as implying that any particular element, step, or function is an essential element that must be included in the claim scope. The scope of patented subject matter is defined only by the claims. Moreover, none of the claims is intended to invoke 35 U.S.C. § 112(f) unless the exact words “means for” are followed by a participle. Use of any other term, including without limitation “mechanism,” “module,” “device,” “unit,” “component,” “element,” “member,” “apparatus,” “machine,” “system,” “processor,” or “controller,” within a claim is understood by the Applicant to refer to structures known to those skilled in the relevant art and is not intended to invoke 35 U.S.C. § 112(f).

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present disclosure and its advantages, reference is now made to the following description taken in conjunction with the accompanying drawings, in which like reference numerals represent like parts:

(2) FIG. 1 illustrates an example network configuration including an electronic device in accordance with this disclosure;

(3) FIGS. 2A and 2B illustrate an example camera configuration on a head mounted device (HMD) for video see-through (VST) augmented reality (AR) in accordance with this disclosure;

(4) FIGS. 3A and 3B illustrate an example process of video rendering for VST AR in accordance with this disclosure;

(5) FIG. 4 illustrates an example video generation pipeline for VST AR in accordance with this disclosure;

(6) FIG. 5 illustrates an example video rendering for VST AR in accordance with this disclosure;

(7) FIG. 6 illustrates an example virtual view computation for VST AR in accordance with this disclosure;

(8) FIG. 7 illustrates an example camera layout for VST AR in accordance with this disclosure;

(9) FIG. 8 illustrates an example extension of video rendering for VST AR in accordance with this disclosure; and

(10) FIG. 9 illustrates an example method for video rendering for VST AR according to this disclosure.

### DETAILED DESCRIPTION

(11) FIGS. 1 through 9, described below, and the various embodiments of this disclosure are described with reference to the accompanying drawings. However, it should be appreciated that this disclosure is not limited to these embodiments, and all changes and/or equivalents or replacements thereto also belong to the scope of this disclosure. The same or similar reference denotations may be used to refer to the same or similar elements throughout the specification and the drawings.

(12) As noted above, augmented reality (AR) systems can seamlessly blend virtual objects generated by computer graphics within real-world scenes. Optical see-through (OST) AR systems refer to AR systems in which users directly view real-world scenes through head-mounted devices (HMDs). Unfortunately, OST AR systems face many challenges that can limit their adoption. Some of these challenges include limited fields of view, limited usage spaces (such as indoor-only usage),

failure to display fully-opaque black objects, and usage of complicated optical pipelines that may require projectors, waveguides, and other optical elements.

(13) In contrast to OST AR systems, video see-through (VST) AR systems present users with generated video sequences of real-world scenes. VST AR systems can be built using virtual reality (VR) technologies and can have various advantages over OST AR systems. For example, VST AR systems can provide wider fields of view and can provide improved contextual augmented reality. Compared to OST AR pipelines, a VST AR pipeline can create virtual views from views of real cameras (referred to as “see-through” cameras), which may require substantial computational resources. This disclosure provides methods and devices for video rendering for VST AR. Among other things, these methods and devices can reduce latency of a virtual view creation pipeline, which can enhance user experiences. Moreover, these methods and devices can produce high-quality virtual views, which can be important based on a user's ability to easily compare generated virtual views with real views.

(14) FIG. 1 illustrates an example network configuration **100** including an electronic device in accordance with this disclosure. The embodiment of the network configuration **100** shown in FIG. 1 is for illustration only. Other embodiments of the network configuration **100** could be used without departing from the scope of this disclosure.

(15) According to embodiments of this disclosure, an electronic device **101** is included in the network configuration **100**. The electronic device **101** can include at least one of a bus **110**, a processor **120**, a memory **130**, an input/output (I/O) interface **150**, a display **160**, a communication interface **170**, and a sensor **180**. In some embodiments, the electronic device **101** may exclude at least one of these components or may add at least one other component. The bus **110** includes a circuit for connecting the components **120-180** with one another and for transferring communications (such as control messages and/or data) between the components.

(16) The processor **120** includes one or more processing devices, such as one or more microprocessors, microcontrollers, digital signal processors (DSPs), application specific integrated circuits (ASICs), or field programmable gate arrays (FPGAs). In some embodiments, the processor **120** includes one or more of a central processing unit (CPU), an application processor (AP), a communication processor (CP), or a graphics processor unit (GPU). The processor **120** is able to perform control on at least one of the other components of the electronic device **101** and/or perform an operation or data processing relating to communication or other functions. As described below, the processor **120** may perform one or more functions related to video rendering for VST AR.

(17) The memory **130** can include a volatile and/or non-volatile memory. For example, the memory **130** can store commands or data related to at least one other component of the electronic device **101**. According to embodiments of this disclosure, the memory **130** can store software and/or a program **140**. The program **140** includes, for example, a kernel **141**, middleware **143**, an application programming interface (API) **145**, and/or an application program (or “application”) **147**. At least a portion of the kernel **141**, middleware **143**, or API **145** may be denoted an operating system (OS).

(18) The kernel **141** can control or manage system resources (such as the bus **110**, processor **120**, or memory **130**) used to perform operations or functions implemented in other programs (such as the middleware **143**, API **145**, or application **147**). The kernel **141** provides an interface that allows the middleware **143**, the API **145**, or the application **147** to access the individual components of the electronic device **101** to control or manage the system resources. The application **147** may include one or more applications that, among other things, perform one or more functions related to video rendering for VST AR. These functions can be performed by a single application or by multiple applications that each carries out one or more of these functions. The middleware **143** can function as a relay to allow the API **145** or the application **147** to communicate data with the kernel **141**, for instance. A plurality of applications **147** can be provided. The middleware **143** is able to control work requests received from the applications **147**, such as by allocating the priority of using the

system resources of the electronic device **101** (like the bus **110**, the processor **120**, or the memory **130**) to at least one of the plurality of applications **147**. The API **145** is an interface allowing the application **147** to control functions provided from the kernel **141** or the middleware **143**. For example, the API **145** includes at least one interface or function (such as a command) for filing control, window control, image processing, or text control.

(19) The I/O interface **150** serves as an interface that can, for example, transfer commands or data input from a user or other external devices to other component(s) of the electronic device **101**. The I/O interface **150** can also output commands or data received from other component(s) of the electronic device **101** to the user or the other external device.

(20) The display **160** includes, for example, a liquid crystal display (LCD), a light emitting diode (LED) display, an organic light emitting diode (OLED) display, a quantum-dot light emitting diode (QLED) display, a microelectromechanical systems (MEMS) display, or an electronic paper display. The display **160** can also be a depth-aware display, such as a multi-focal display. The display **160** is able to display, for example, various contents (such as text, images, videos, icons, or symbols) to the user. The display **160** can include a touchscreen and may receive, for example, a touch, gesture, proximity, or hovering input using an electronic pen or a body portion of the user.

(21) The communication interface **170**, for example, is able to set up communication between the electronic device **101** and an external electronic device (such as a first electronic device **102**, a second electronic device **104**, or a server **106**). For example, the communication interface **170** can be connected with a network **162** or **164** through wireless or wired communication to communicate with the external electronic device. The communication interface **170** can be a wired or wireless transceiver or any other component for transmitting and receiving signals.

(22) The wireless communication is able to use at least one of, for example, long term evolution (LTE), long term evolution-advanced (LTE-A), 5th generation wireless system (5G), millimeter-wave or 60 GHz wireless communication, Wireless USB, code division multiple access (CDMA), wideband code division multiple access (WCDMA), universal mobile telecommunication system (UMTS), wireless broadband (WiBro), or global system for mobile communication (GSM), as a cellular communication protocol. The wired connection can include, for example, at least one of a universal serial bus (USB), high definition multimedia interface (HDMI), recommended standard 232 (RS-232), or plain old telephone service (POTS). The network **162** or **164** includes at least one communication network, such as a computer network (like a local area network (LAN) or wide area network (WAN)), Internet, or a telephone network.

(23) The electronic device **101** further includes one or more sensors **180** that can meter a physical quantity or detect an activation state of the electronic device **101** and convert metered or detected information into an electrical signal. For example, the sensor(s) **180** can include one or more cameras or other imaging sensors, which may be used to capture images of scenes. The sensor(s) **180** can also include one or more buttons for touch input, one or more microphones, a gesture sensor, a gyroscope or gyro sensor, an air pressure sensor, a magnetic sensor or magnetometer, an acceleration sensor or accelerometer, a grip sensor, a proximity sensor, a color sensor (such as a red green blue (RGB) sensor), a bio-physical sensor, a temperature sensor, a humidity sensor, an illumination sensor, an ultraviolet (UV) sensor, an electromyography (EMG) sensor, an electroencephalogram (EEG) sensor, an electrocardiogram (ECG) sensor, an infrared (IR) sensor, an ultrasound sensor, an iris sensor, or a fingerprint sensor. The sensor(s) **180** can further include an inertial measurement unit, which can include one or more accelerometers, gyroscopes, and other components. In addition, the sensor(s) **180** can include a control circuit for controlling at least one of the sensors included here. Any of these sensor(s) **180** can be located within the electronic device **101**.

(24) The first external electronic device **102** or the second external electronic device **104** can be a wearable device or an electronic device-mountable wearable device (such as an HMD). When the electronic device **101** is mounted in the electronic device **102** (such as the HMD), the electronic



device **101** can communicate with the electronic device **102** through the communication interface **170**. The electronic device **101** can be directly connected with the electronic device **102** to communicate with the electronic device **102** without involving with a separate network. The electronic device **101** can also be an augmented reality wearable device, such as eyeglasses, that include one or more cameras.

(25) The first and second external electronic devices **102** and **104** and the server **106** each can be a device of the same or a different type from the electronic device **101**. According to certain embodiments of this disclosure, the server **106** includes a group of one or more servers. Also, according to certain embodiments of this disclosure, all or some of the operations executed on the electronic device **101** can be executed on another or multiple other electronic devices (such as the electronic devices **102** and **104** or server **106**). Further, according to certain embodiments of this disclosure, when the electronic device **101** should perform some function or service automatically or at a request, the electronic device **101**, instead of executing the function or service on its own or additionally, can request another device (such as electronic devices **102** and **104** or server **106**) to perform at least some functions associated therewith. The other electronic device (such as electronic devices **102** and **104** or server **106**) is able to execute the requested functions or additional functions and transfer a result of the execution to the electronic device **101**. The electronic device **101** can provide a requested function or service by processing the received result as it is or additionally. To that end, a cloud computing, distributed computing, or client-server computing technique may be used, for example. While FIG. **1** shows that the electronic device **101** includes the communication interface **170** to communicate with the external electronic device **104** or server **106** via the network **162** or **164**, the electronic device **101** may be independently operated without a separate communication function according to some embodiments of this disclosure.

(26) The server **106** can include the same or similar components as the electronic device **101** (or a suitable subset thereof). The server **106** can support to drive the electronic device **101** by performing at least one of operations (or functions) implemented on the electronic device **101**. For example, the server **106** can include a processing module or processor that may support the processor **120** implemented in the electronic device **101**. As described below, the server **106** may perform one or more functions related to video rendering for VST AR.

(27) Although FIG. **1** illustrates one example of a network configuration **100** including an electronic device **101**, various changes may be made to FIG. **1**. For example, the network configuration **100** could include any number of each component in any suitable arrangement. In general, computing and communication systems come in a wide variety of configurations, and FIG. **1** does not limit the scope of this disclosure to any particular configuration. Also, while FIG. **1** illustrates one operational environment in which various features disclosed in this patent document can be used, these features could be used in any other suitable system.

(28) FIGS. **2A** and **2B** illustrate an example camera configuration **200** on an HMD **202** for VST AR in accordance with this disclosure. For ease of explanation, the HMD **202** may be described as being implemented using the electronic device **101** in the network configuration **100** of FIG. **1**. However, the HMD **202** may be used in any other suitable device(s) and in any other suitable system(s).

(29) As shown in FIGS. **2A** and **2B**, the HMD **202** can be used for VST AR. In this example, the HMD **202** can include one or more see-through cameras **204** and at least first and second lenses **206**. The HMD **202** can have rendering viewpoints **208** based on locations of the user's eyes when the HMD **202** is worn. Each rendering viewpoint **208** can also correspond to a virtual camera viewpoint for a virtual camera that is discussed later in regard to a virtual camera pose. A virtual camera pose is a pose for a "virtual camera" corresponding to a pose of the user's eye. Each rendering viewpoint **208** can be determined based on measurements taken once a user is wearing the HMD **202** or can have a default standard position (and the user may be able to adjust the lenses **206** or the fit of the HMD **202**).

(30) In some cases, the see-through cameras **204** can be arranged on an exterior of the HMD **202**. When multiple cameras **204** are present, the cameras **204** can be positioned at different distances and orientations in relation to the eye viewpoints and the rendering viewpoints **208**. For example, a first camera **204a** can have a distance in the x-direction or horizontal distance **210** from a rendering viewpoint **208**, a distance in the y-direction or vertical distance **212** from that rendering viewpoint **208**, and a distance in the z-direction or depth **214** from that rendering viewpoint **208**.

(31) Although FIGS. 2A and 2B illustrate one example of a camera configuration **200** on an HMD **202** for VST AR, various changes may be made to FIGS. 2A and 2B. For example, an HMD **202** may have any other suitable form factor, and the number and placement of the cameras **204** can vary as needed or desired. Also, the HMD **202** may be used in any other suitable video rendering process and is not limited to the specific processes described above.

(32) FIGS. 3A and 3B illustrate an example process **300** of video rendering for VST AR in accordance with this disclosure. For ease of explanation, the process **300** of FIGS. 3A and 3B is described as being performed using the electronic device **101** in the network configuration **100** of FIG. 1, where the electronic device **101** may be implemented as the HMD **202** of FIG. 2. However, the process **300** may be used with any other suitable device(s) and in any other suitable system(s).

(33) As shown in FIGS. 3A and 3B, the process **300** includes an image and depth data operation **302**, a virtual view synthesis operation **304**, and a virtual view rendering operation **306**. The process **300** can capture data from each of the cameras **204** (and optionally one or more other sensors of the electronic device **101**) and output a see-through display with virtual components combined with a real-scene captured by the cameras **204**. The process **300** here can determine a contribution factor for each camera and merge the contributions from the cameras into a combined camera view.

(34) In this example, the image and depth data operation **302** can include an identify camera operation **308**. The identify camera operation **308** can determine or select a camera C, **204** for use when performing additional operations **310-322**. In some cases, the electronic device **101** can include a single camera **204**, and the identify camera operation **308** may be omitted. In other cases, the electronic device **101** can include multiple cameras **204**, such as when the electronic device **101** includes one or more sensors **180** that implement the cameras **204**. In particular embodiments, the cameras **204** can be arranged in a camera configuration **200** or in another pattern to obtain a desired or optimized field of view (FOV) of the HMD **202**. The operations **310-322** can be performed for each camera **204** in the camera configuration **200** on the HMD **202**.

(35) The electronic device **101** can perform an image capture operation **310** in order to capture an image using the selected camera **204**. Here, the selected camera **204** can be used to capture an image I.sub.i, where i represents the i.sup.th selected camera. Having multiple cameras **204** in a camera arrangement allows for images to be captured at different positions and orientations in order to obtain multiple views of objects in a scene. This allows for more realistic rendering of objects and also a more aesthetic relationship with virtual objects. The electronic device **101** can also perform a camera pose operation **312** in order to generate position data related to the current image I.sub.i for the current camera C.sub.i. The position data can identify a position and an orientation of the selected camera **204**. In some embodiments, the position and orientation of the selected camera **204** can be fixed in relation to an IMU position sensor.

(36) The electronic device **101** can further perform a camera calibration operation **314** in order to determine intrinsic and extrinsic parameters of the cameras **204**. The camera calibration operation **314** can use any suitable technique to identify the intrinsic and extrinsic parameters of the cameras **204**. The intrinsic and extrinsic parameters can be used with the current image in order to perform an image un-distortion operation **316**, which can be performed to un-distort the current image. For example, while wide-angle cameras can capture images at wide views, edge distortion due to lens shape is almost unavoidable. The image un-distortion operation **316** can use any suitable techniques to un-distort images. For instance, the image un-distortion operation **316** may use the

intrinsic parameters and distortion model of the selected camera **204** to un-distort the current image.

(37) The electronic device **101** can also perform an image rectification operation **318** to the current image so that the image data is not raw and is ready for feature matching. The rectification is performed to transform images from multiple cameras (or images from a single camera) and therefore multiple angles into a common plane to simplify the feature matching process. The result is that the multiple images are now each viewed from the same perspective or angle so that the images can be placed in the same common plane, which allows the points in the images to be much more easily matched (such as by using parallel matching lines). A dense depth map operation **320** can be performed to generate a dense depth map for the current image  $I_{sub.i}$ . Depths in a scene can be reconstructed based on the current image from the selected camera **204** using the rectified image generated by the image rectification operation **318**. The dense depth map operation **320** can therefore be performed to generate a dense depth map for the current image based on a respective rectified image. In some cases, the dense depth map operation **320** can receive a current frame and output depth information corresponding to pixel points in the frame. A dense depth map can be used for various functions, such as occlusion between virtual objects and a real-world scene, depth matching between perceptual depths and real-world scene depths, and depth re-projection in viewpoint transformations.

(38) A virtual camera pose operation **322** can determine a virtual camera pose  $S$ , for use with the virtual view synthesis operation **304**. As discussed previously in relation to FIGS. 2A and 2B, a virtual camera can be associated with each of the user's eyes and the rendering viewpoints **208**. The virtual camera pose operation **322** can identify a location and orientation of each eye of the user or each rendering viewpoint **208** in order to determine a virtual camera pose. Again, note that the operations **310-322** can be performed for each camera **204** in order to process a collection of images captured using the cameras **204**.

(39) The virtual view synthesis operation **304** can reconstruct the viewpoints of the virtual cameras for the left and right eyes of a user. As shown in FIG. 3B, the virtual view synthesis operation **304** includes input operations **324**, a virtual view projection operation **326**, a virtual view depth map projection operation **328**, a parallax operation **330**, a contribution and blending operation **332**, a camera complete determination **334**, an un-occlusion operation **336**, a reverse transform operation **338**, and a views complete determination **340**. The input operation **324** includes a current image operation **342**, a current depth map operation **344**, and a calculate virtual view operation **346**. The input operation **324** can determine an order for processing the cameras **204** through operations **326-334**. In some cases, the electronic device **101** can receive a current image for the selected camera **204** in the processing order for the current image operation **342** and a current depth map for the selected camera **204** in the processing order for the current depth map operation **344**. For example, the current image can be received as a result of the image rectification operation **318**, and the current depth map can be received as a result of the dense depth map operation **320**. The electronic device **101** can track a progress of processing each camera **204** through the operations **324-334**.

(40) The calculate virtual view operation **346** can determine virtual viewpoints  $V_{sub.j}$  of the virtual cameras for adjusting each of the current images. In some cases, the calculation can utilize the current camera pose  $S_{sub.i}$  and the virtual camera pose  $S_{sub.j}$  to determine the virtual viewpoints. In some embodiments, the subscript  $j$  here can correspond to or indicate either left or right, such as a left virtual camera viewpoint  $V_{sub.left}$  or a right virtual camera viewpoint  $V_{sub.right}$ . The viewpoints are based on the current image  $I_{sub.i}$  and the current virtual viewpoint  $V_{sub.j}$ . For example, the calculate virtual view operation **346** can determine how to perform warping in order to transfer one or more video frames from see-through camera positions to virtual camera positions, which can be used to generate the left virtual camera viewpoint  $V_{sub.left}$  and the right virtual camera viewpoint  $V_{sub.right}$ . As a particular example, the calculate virtual view operation **346** may generate one or more transformations that help match camera viewpoints to virtual viewpoints

V.sub.j.

(41) The virtual view depth map projection operation **328** can use the current dense depth maps to perform depth re-projections. For instance, a current dense depth map can be depth re-projected from a camera viewpoint to the left virtual camera viewpoint V.sub.left and the right virtual camera viewpoint V.sub.right. The depths are adjusted based on the difference between the camera viewpoint and the virtual viewpoint. The parallax operation **330** can be performed to identify adjustments that might be needed to remove view artifacts in an overlapping area between the cameras. For example, because a camera viewpoint is not identical to eye and rendering viewpoints, objects at different depths can be distorted if adjusted equally. As a particular example, objects closer in depths to the viewpoints may be adjusted at greater rates than objects at further distances or a background of a real-world scene. The parallax operation **330** can determine differences in viewing relative pixel locations or objects in three-dimensional (3D) space based on line-of-sight and can correct these differences in virtual views. For instance, the parallax operations **330** may correct pixels or objects in the real-world scene that are affected by the parallax in three frames captured by three different cameras.

(42) The contribution and blending operation **332** can be used to determine contributions of each image from a camera to be blended into a virtual view. For example, the contribution and blending operation **332** can determine a contribution for each pixel from a specific camera for a scene using a parallax map P.sub.i. The contribution can be based on a confidence of the pixel from a camera. For the contribution and blending operation **332**, the electronic device **101** can respectively combine the pixel contributions from each image into virtual video frames. A location in the virtual view for each pixel in an image can be identified using the parallax information calculated by the parallax operation **330**. For the camera complete determination **334**, the electronic device **101** can determine whether each camera's contributions have been blended into the virtual video frames. The operations **324-332** here can be repeated until the contributions of all see-through cameras **204** are computed and blended with the virtual view contribution and blending operation **332**. When at least one of the cameras **204** has not been processed, the electronic device returns to the input operation **324**.

(43) When all of the cameras **204** have been processed, the electronic device **101** can perform an un-occlusion operation **336** on the virtual video frame. The un-occlusion operation **336** can detect missing or incorrect data that develops during the contribution and blending operation **332**. Holes or missing information can be introduced when occlusions are created by depth warping. The missing information can be filled in various ways, such as pixel expansion, pixel patch replacement from previous image captures, or any other suitable technique for recovering missing information. The reverse transform operation **338** transforms the virtual video frame to fit a display or multiple displays. The reverse transform operation **338** can be performed to identify and compensate for lens distortions of a VST headset. For example, the reverse transform operation **338** can be used to identify any geometric distortions in a warped image and to remap a source image to an un-warped image as if it was taken with a perspective lens. The reverse transform operation **338** can also calibrate an image based on known different indices of refraction across a lens, which can increase false colors in an image as distance for a pixel is further from a center of the image. Any chromatic aberrations can be corrected according to lens data and focal length information using the reverse transform operation **338**, where the electronic device **101** uses the information from the reverse transform operation **338** to correct the virtual view video frame based on characteristics of first and second display panels (which are described below).

(44) For the views complete determination **340**, the electronic device **101** can determine whether a left virtual view V.sub.left **348** and a right virtual V.sub.right **350** were generated. The electronic device **101** can repeat the operations **324-338** until all virtual views have been generated. At this point, the electronic device **101** prepares the left virtual view V.sub.left **348** and the right virtual view V.sub.right **350** for virtual view rendering operations **306**. When one of the views has not

been generated, the electronic device **101** performs the input operation **324** for the non-generated view and resets any counters or tracking related to the camera complete determination **334**.

(45) As shown in FIG. **3A**, the virtual view rendering operation **306** can perform post-processing on the left virtual view **348** and the right virtual view **350**. For the virtual view rendering operations **306**, the electronic device **101** can blend virtual information **352** with the left virtual view V.sub.left **348** and the right virtual view V.sub.right **350** in blending operations **356** and **358**. For the blending operations **356** and **358**, the electronic device **101** can respectively combine virtual objects into the left virtual view V.sub.left **348** and the right virtual view V.sub.right **350**. The virtual objects can be positioned, oriented, and colored based on the information from the virtual information **352**. In some cases, one or more of these operations can be implemented using processing on a GPU with CPU/GPU interoperability to share memory buffers between the CPU and the GPU. The blending operations **356** and **358** can use the first and second dense depth maps for occlusion between virtual objects and the real-world scene. First and second display rendering operations **360** and **362** can render the left virtual view V.sub.left **348** and the right virtual view V.sub.right **350** for display on a display panel. In some cases, the electronic device **101** can display a first video frame on a first panel and a second video frame on a second panel. Depending on the implementation, the left virtual view V.sub.left **348** and the right virtual view V.sub.right **350** can be pre-rendered or can be dynamically rendered.

(46) Although FIGS. **3A** and **3B** illustrate one example of a process **300** of video rendering for VST AR, various changes may be made to FIGS. **3A** and **3B**. For example, while shown as a series of operations, various operations in FIGS. **3A** and **3B** may overlap, occur in parallel, occur in a different order, or occur any number of times.

(47) FIG. **4** illustrates an example VST AR pipeline **400** in accordance with this disclosure. For ease of explanation, the VST AR pipeline **400** may be described as being implemented using the electronic device **101** in the network configuration **100** of FIG. **1**, where the electronic device **101** may be implemented as the HMD **202** of FIG. **2**. However, the VST AR pipeline **400** may be used in any other suitable device(s) and in any other suitable system(s).

(48) As shown in FIG. **4**, the VST AR pipeline **400** can be used to generate and present one or more AR or VR objects on two panels **402**, where the panels **402** are viewed by a user's eyes **404** through dedicated lenses **406**. In some embodiments, the panels **402** can be VST AR displays that are see-through with the exception of one or more projected AR or VR objects on the panels **402**. Any suitable AR or VR objects may be projected onto the panels **402**. In some embodiments, one or more AR or VR objects may be selected for presentation based on the real-world scene being viewed by the user or an application being executed on a device implementing the VST AR pipeline **400**.

(49) The VST AR pipeline **400** also incorporates a blender **408** that can receive real-world information **410** and virtual information **412** and that can process the information **410** and **412** in order to generate one or more AR or VR objects for display on the panels **402**. In some cases, the real-world information **410** can include information captured from one or more sensors, such as one or more optical sensors, accelerometers, gravity sensors, ambient light sensors, proximity sensors, magnetism sensors, gyroscopes, position sensors, etc. Also, in some cases, the virtual information **412** can include information related to the one or more AR or VR objects to be presented to the user, and different virtual information **412** may be associated with different real-world information **410**. The virtual information **412** may be stored on a device implementing the VST AR pipeline **400** or on another device, such as a server **106**. In some embodiments, the blender **408** can represent at least one processing device, such as the processor **120**. In this particular example, the blender **408** is shown as including a CPU or GPU, although other implementations of the blender **408** are possible using other types of processing devices.

(50) Although FIG. **4** illustrates one example of a VST AR pipeline **400**, various changes may be made to FIG. **4**. For example, the number of various components of the pipeline **400** can vary as

needed or desired. As particular examples, the VST AR pipeline **400** may include multiple blenders **408**, such as one blender **408** per panel **402** or multiple blenders **408** per panel **402**. In addition, the VST AR pipeline **400** may be used to perform any suitable video transformation process.

(51) FIG. 5 illustrates an example technique **500** for video rendering of VST AR in accordance with this disclosure. For ease of explanation, the technique **500** of FIG. 5 is described as being performed using the electronic device **101** in the network configuration **100** of FIG. 1, where the electronic device **101** may be implemented as the HMD **202** of FIG. 2. However, the technique **500** may be used with any other suitable device(s) and in any other suitable system(s).

(52) As shown in FIG. 5, the technique **500** uses three see-through cameras **204a-204c** but can be modified for use with any suitable number of cameras **204**. The see-through cameras **204a-204c** can capture images **502a-502c** of an object **504**. Rectified images **506a-506c** can be respectively obtained from the images **502a-502c**, such as when the rectified images **506a-506c** are generated using the image rectification operation **318**. Corresponding epipolar lines of the rectified images **506a-506c** are co-linear and can be used as stereo pairs.

(53) Depths **508a-508c** can be computed as distances from the cameras **204a-204c** to the object **504**. In embodiments with a single camera **204**, a deep learning approach or other suitable technique can be utilized to estimate a depth map for the single camera **204**. In embodiments with multiple cameras **204**, stereo camera pairs or other techniques can be utilized to compute depth maps for each camera **204a-204c**. In some cases, a deep neural network (DNN) or other machine learning model can be used to extract feature maps and to estimate disparity maps. A dense depth map can be generated from the extracted feature maps, estimated disparity maps, and computed depth maps. The dense depths maps can be generated according to the dense depth map operation **320**.

(54) Contributions, contribution confidence maps, parallax maps, and parallax contribution maps can be generated for each camera **204a-204c** in the parallax operation **330** and the contributions and blend operation **332**. The contributions, contribution confidence maps, parallax maps, and parallax confidence maps can be used to create and render the left virtual view V.sub.left **348** for the left virtual camera **510a** and the right virtual view V.sub.right **350** for the right virtual camera **510b** in the left and right rendering operations **360** and **362**.

(55) Although FIG. 5 illustrates one example of a technique **500** for video rendering of VST AR, various changes may be made to FIG. 5. For example, while shown as a series of operations, various operations in FIG. 5 may overlap, occur in parallel, occur in a different order, or occur any number of times.

(56) FIG. 6 illustrates an example system **600** for virtual view computation for VSTAR in accordance with this disclosure. For ease of explanation, the system **600** may be described as being implemented using the electronic device **101** shown in FIG. 1, the HMD **202** shown in FIG. 2, or the blender **408** shown in FIG. 4. However, the system **600** may be used in any other suitable device(s) and in any other suitable system(s).

(57) As shown in FIG. 6, the system **600** can produce a final left virtual view **348** and a final right virtual view **350**. The system **600** can include an input module **602**, a contribution module **604**, an integration module **606**, and a post-processing module **608**. Depending on the implementation, the system **600** may be implemented in an HMD or other immersive reality device, and the processor **120** may be used to perform the functions for each of the input module **602**, the contribution module **604**, the integration module **606**, and the post-processing module **608**.

(58) In this example, the input module **602** can receive, capture, store, and process the inputs for the system **600**. For example, the input module **602** can prepare images **610a-610c** and corresponding dense depth maps **612a-612c**. While the illustrative example shows the system **600** including three cameras **204**, the operations can be performed in systems with more or less than three cameras **204**. The electronic device **101** can perform operations **308-318** to prepare the images **610a-610c** and can perform the dense depth map operation **320** to prepare the depth maps

**612a-612c** that correspond to the images **610a-610c**.

(59) The contribution module **604** can determine a factor or contribution **614a-614c** for each image **610a-610c** used for the virtual left view **348** and the virtual right view **350**. For example, the contribution module **604** can compute the contributions **614a-614c** in the contribution and blending operation **332**. The contribution module **604** can compute a contribution **614a-614c** of a specified see-through camera **204a-204c** to a virtual view of a virtual camera and compute contribution confidence maps **618a-618c** for respective contributions **614a-614c**. The contribution confidence maps **618a-618c** can provide a ratio or factor for how much of the contribution **614a-614c** is used in generating the left and right virtual views **348** and **350**. A high confidence indicates a high similarity in pose between the see-through camera **204** and the virtual camera, and low confidence indicates a low similarity in pose between the see-through camera **204** and the virtual camera. The operations of the contribution module **604** may be performed separately for the left virtual view **348** and the right virtual view **350**. The contribution module **604** can also compute a parallax map **616a-616c** of each see-through camera **204** to the virtual view of a virtual camera using the depth maps **612a-612c**. The parallax map **616a-616c** can be generated when the electronic device **101** performs the parallax operation **330**. The parallax maps **616a-616c** can identify displacement of objects at different depths. The parallax confidence maps **620a-620c** are similar to the contribution confidence maps **618a-618c**, but the parallax confidence maps **620a-620c** are directed to a confidence of the parallax maps **616a-616c**.

(60) The integration module **606** can integrate the contributions **614a-614c**, the parallax maps **616a-616c**, the contribution confidence maps **618a-618c**, and the parallax confidence maps **620a-620c** to generate rough left and right virtual views **620a** and **620b**. For example, the integration module **606** can generate the rough left and right virtual views **620a** and **620b** by performing operation **332**. The integration module **606** can integrate the contributions **614a-614c** and parallax maps **616a-616c** by using a contribution **614** with a higher confidence on a front layer and a lower confidence as a back layer for the rough left and right virtual views **620a** and **620b**. The use of the confidence maps can reduce artifacts of mis-occlusion and mixing of foreground and background color at a discontinuity of a depth map. The post-processing module **608** can adjust the rough left and right virtual views **620a** and **620b** to correct any deficiencies detected during the integration process. For instance, the rough left and right virtual views **620a** and **620b** can have some artifacts from the integration process that includes un-occlusion, object contours, sampling gaps, etc. The post-processing module **608** can recover missing information from an image frame sequence and hole filling operations.

(61) Although FIG. **6** illustrates one example of a system **600** for virtual view computation for VST AR, various changes may be made to FIG. **6**. For example, the system **600** may have dedicated input modules and contribution modules for each camera or more than one for each of the input module **602**, the contribution module **604**, the integration module **606**, and the post-processing module **608**. In addition, the system **600** may be used in any other suitable video rendering device and is not limited to the specific processes described above.

(62) FIG. **7** illustrates an example camera layout **700** in VST AR in accordance with this disclosure. For ease of explanation, the camera layout **700** of FIG. **7** is described as being used with the electronic device **101** in the network configuration **100** of FIG. **1**, where the electronic device **101** may be implemented as the HMD **202** of FIG. **2**. However, the camera layout **700** may be used with any other suitable device(s) and in any other suitable system(s). The camera layout **700** can be used for explaining a creation of one or more confidence maps.

(63) As shown in FIG. **7**, the camera layout **700** illustrates a relationship between a see-through camera **204** and a virtual camera **701** at an eye viewpoint. The relationship discussed below can be extend to each camera **204** on an HND **202**. For the camera layout **700**, a pose of the camera **204** is defined by an x-rotation ( $\phi_{\text{sub.ix}}$ ) **702**, a y-rotation ( $\phi_{\text{sub.iy}}$ ) **704**, and a z-rotation ( $\phi_{\text{sub.iz}}$ ) **706**. Also, a pose of the virtual camera **701** is defined based on an x-rotation ( $\alpha_{\text{sub.x}}$ ) **708**, a y-rotation

( $\alpha$ .sub.y) **710**, and a z-rotation ( $\alpha$ .sub.z) **712**. In addition, a distance between the camera **204** and the virtual camera **701** is defined based on a horizontal distance ( $d$ .sub.ix) **210**, a vertical distance ( $d$ .sub.iy) **212**, and a depth ( $d$ .sub.iz) **214**. In some cases, an orientation difference and a position difference between the see-through camera **204** and the virtual camera **701** can be determined as follows.

$$\text{diff.sub.oi} = \sqrt{(\alpha.\text{sub.x} - \phi.\text{sub.ix}).\text{sup.2} + (\alpha.\text{sub.y} - \phi.\text{sub.iy}).\text{sup.2} + (\alpha.\text{sub.z} - \phi.\text{sub.iz}).\text{sup.2}} \quad (1)$$

$$\text{diff.sub.pi} = \sqrt{(d.\text{sub.ix}.\text{sup.2} + d.\text{sub.iy}.\text{sup.2} + d.\text{sub.iz}.\text{sup.2})} \quad (2)$$

Here,  $\text{diff.sub.oi}$  represents an orientation difference, and  $\text{diff.sub.pi}$  represents a position difference. A total orientation difference and a total position difference for all the see-through cameras **204** can be determined as follows.

$$\text{diff.sub.o} = \sum_{i=1}^n \text{diff.sub.oi} \quad (3)$$

$$\text{diff.sub.p} = \sum_{i=1}^n \text{diff.sub.pi} \quad (4)$$

Here,  $\text{diff.sub.o}$  represents a total orientation difference, and  $\text{diff.sub.p}$  represents a total position difference. In some cases, the total differences can be normalized as follows.

$$(64) \quad \text{diff}_{\text{oni}} = \frac{\text{diff}_{\text{oi}}}{\text{diff}_{\text{o}}}, (i = 1, 2, \dots, n) \quad (5) \quad \text{diff}_{\text{pni}} = \frac{\text{diff}_{\text{pi}}}{\text{diff}_{\text{p}}}, (i = 1, 2, \dots, n) \quad (6)$$

Here,  $\text{diff.sub.oni}$  represents a normalized orientation difference, and  $\text{diff.sub.pni}$  represents a normalized position difference. The confidence for each see-through camera  $C.\text{sub.i}$  ( $i=1, 2, \dots, n$ ) can be determined as follows.

$$(65) \quad m_i = \frac{1}{\text{oi} \cdot \text{diff}_{\text{oni}}} + \frac{1}{\text{pi} \cdot \text{diff}_{\text{pni}}}, (i = 1, 2, \dots, n) \quad (7)$$

Here,  $m.\text{sub.i}$  represents a confidence of a see-through camera  $C.\text{sub.i}$ , and ( $\lambda.\text{sub.oi}, \lambda.\text{sub.pi}$ ) represent the weights for the information from the orientation difference and the position difference. In some cases, the weights for the orientation difference and the position difference can be determined as follows.

$$\lambda.\text{sub.oi} + \lambda.\text{sub.pi} = 1 \quad (8)$$

(66) The virtual view can be integrated by blending the contributions of all see-through cameras **204** with each corresponding confidence map. In some cases, the color of the virtual view can be determined as follows.

$$c.\text{sub.j} = \sum_{n=1}^n w.\text{sub.n} \cdot c.\text{sub.n} \quad (9)$$

Here,  $c.\text{sub.i}$  represents a color, and  $w.\text{sub.i}$  represents a weight for color blending. In some cases, the weight  $w$ , can be determined as follows.

$$Z.\text{sub.i} = \frac{1}{\sum_{n=1}^n w.\text{sub.n}} \quad (10)$$

(67) A normalized confidence can be used to determine a factor in relation to each of the other cameras. In some cases, the normalized confidence can be determined as follows.

$$(68) \quad m_{\text{ni}} = \frac{m_i}{\sum_{i=1}^n m_i} (i = 1, 2, \dots, n) \quad (11)$$

(69) Here,  $m.\text{sub.ni}$  represents a normalized confidence of a see-through camera  $C.\text{sub.i}$ . When a current point in a field of view is not in view or is obstructed from the view of a see-through camera **204**, the confidence may be set to zero or a nominal value in the confidence map of an image corresponding to the specified see-through camera **204**.

(70) Although FIG. 7 illustrates one example of a camera layout **700** in VST AR, various changes may be made to FIG. 7. For example, the camera layout **700** can include any suitable number of cameras and can be replicated for both eyes of a user and multiple lenses or panels of an HMD **202**. In addition, the camera layout **700** may be used in any other suitable video rendering device and is not limited to the specific processes described above.

(71) FIG. 8 illustrates an example extension **800** of video rendering for VST AR in accordance with this disclosure. The extension **800** shown in FIG. 8 may be used in any of the devices, systems, processes, and techniques described above or below. As shown in FIG. 8, the extension **800** can warp images **802a-802n** into a warped image **804** in 3D space in a global coordinate system with



corresponding depth maps. In some cases, the warped 3D points can be determined as follows.

$$(x1,y1).fwdarw.(x,y,z),(x2,y2).fwdarw.(x,y,z) \quad (12)$$

The warped 3D points can be projected to a virtual view **806** of virtual rendering cameras. In some cases, the projection can be determined as follows.

$$(x,y,z).fwdarw.(x,y) \quad (13)$$

After the virtual views have been projected, post-processing can be performed to fill holes from un-occlusion and sampling gaps for the left and right final views.

(72) In some embodiments, blending operations can determine a see-through camera **204** that is closest in distance to a virtual camera. A virtual view for that virtual camera can be constructed using the image and depth map from the closest see-through camera **204**. One or more additional cameras **204** can be used for post-processing procedures, including filling sampling gaps and filling holes for un-occlusions.

(73) Although FIG. **8** illustrates one example of an extension **800** of video rendering for VST AR, various changes may be made to FIG. **8**. For example, different numbers of cameras can be utilized in the 3D space for the virtual rendering camera. In addition, the extension **800** may be used in any other suitable video rendering device and is not limited to the specific processes described above.

(74) FIG. **9** illustrates an example method **900** for video rendering for VST AR according to this disclosure. For ease of explanation, the method **900** of FIG. **9** is described as being performed using the electronic device **101** of FIG. **1**, where the electronic device **101** may be implemented as the HMD **202** of FIG. **2**. However, the method **900** may be used with any other suitable device(s) and in any other suitable system(s).

(75) As shown in FIG. **9**, the electronic device **101** can capture an image and associated camera pose at operation **902**. For example, an image and associated camera pose can be obtained for each of multiple cameras **204** on an HMD **202**. The electronic device **101** can perform un-distortion operations **316** and image rectification operations **318** on each of the captured images. A depth map can be determined for each image and associated camera pose at operation **904**. For example, the electronic device **101** can perform a dense depth map operation **320** to produce a dense depth map for each captured image, and the depth map can identify depths of pixels in the associated image.

(76) First and second contributions of the captured image can be determined in operation **906**. For example, the electronic device **101** can determine a first contribution of the captured image for a first virtual view for display on a first display of the AR device and a second contribution of the captured image for a second virtual view for display on a second display of the AR device. A distance between each of the cameras to the virtual camera can be determined based on an arrangement of the HMD **202**. First and second contribution confidence maps corresponding to the first and second contributions can be determined in operation **908**. For example, the electronic device **101** can determine a first confidence map for the first virtual view based on the camera pose and a position of the camera in relation to a first virtual camera and a second confidence map for the second virtual view based on the camera pose and the position of the camera in relation to a second virtual camera. The first eye of the user can correspond to the first virtual camera, and the second eye of the user can correspond to the second virtual camera. Confidences for the first and second contribution confidence maps can be determined based on distance and orientation differences between the camera and the respective first and second virtual cameras, where the confidence increases as the difference between the camera and the respective first and second virtual cameras decreases.

(77) First and second parallax maps can be determined from the depth maps at operation **910**. For example, the electronic device **101** may, for each of the cameras **204**, determine a first parallax map for the first virtual camera and a second parallax map for the second virtual camera by projecting a depth map associated with the captured image onto an image plane of the first and second displays. The first parallax map can be used to identify a location of the first contribution in the first virtual view, and the second parallax map can be used to identify a location of the second contribution in

the second virtual view. First and second parallax confidence maps can be determined from the depth map and parallax maps at operation **912**. For example, the electronic device **101** may, for each of the cameras **204**, determine a first parallax confidence map associated with the first parallax map for the first virtual view based on the camera pose and the position of the camera in relation to the first virtual camera and a second parallax confidence map associated with the second parallax map for the second virtual view based on the camera pose and the position of the camera in relation to the second virtual camera. The first and second parallax confidence maps can be used to identify a priority of respective first and second parallax maps in generating the first and second virtual views, where a parallax map with a higher priority may be used in a layer in front of a parallax map with a lower priority.

(78) First and second virtual view can be generated from the contributions and the contribution confidence maps at operation **914**. For example, the electronic device **101** can generate the first virtual view by combining the first contribution using the first confidence map for each of the cameras **204** and the second virtual view by combining the second contribution using the second confidence map for each of the cameras **204**. The first parallax map and the first parallax confidence map can be used in generating the first virtual view, and the second parallax map and the second parallax confidence map can be used in generating the second virtual view.

(79) The first and second virtual views can be displayed in operation **916**. For example, the electronic device **101** can display the first virtual view on a first display and the second virtual view on a second display. As a particular example, the first and second virtual views **348** and **350** can be displayed on the left and right panels **402** shown in FIG. **4** or displayed on the HMD **202** shown in FIG. **2**.

(80) Although FIG. **9** illustrates one example of a method **900** for video rendering for VST AR, various changes may be made to FIG. **9**. For example, while shown as a series of steps, various steps in FIG. **9** may overlap, occur in parallel, occur in a different order, or occur any number of times.

(81) Although this disclosure has been described with example embodiments, various changes and modifications may be suggested to one skilled in the art. It is intended that this disclosure encompass such changes and modifications as fall within the scope of the appended claims.

## Claims

1. A method for video rendering for video see-through (VST) augmented reality (AR) on an AR device, the method comprising: for each of a plurality of cameras of the AR device: capturing an image and associating the captured image with a camera pose; determining (i) a first contribution of the captured image for a first virtual view for display on a first display of the AR device and (ii) a second contribution of the captured image for a second virtual view for display on a second display of the AR device; and determining (i) a first confidence map for the first virtual view based on the camera pose and a position of the camera in relation to a first virtual camera and (ii) a second confidence map for the second virtual view based on the camera pose and the position of the camera in relation to a second virtual camera; and generating (i) the first virtual view by combining the first contribution using the first confidence map for each of the plurality of cameras and (ii) the second virtual view by combining the second contribution using the second confidence map for each of the plurality of cameras.

2. The method of claim 1, further comprising: for each of the plurality of cameras, determining a first parallax map for the first virtual camera and a second parallax map for the second virtual camera by projecting a depth map associated with the captured image onto image planes of the first and second displays; wherein the first parallax map is used in generating the first virtual view and the second parallax map is used in generating the second virtual view.

3. The method of claim 2, wherein: the first parallax map is used to identify a location of the first

contribution in the first virtual view; and the second parallax map is used to identify a location of the second contribution in the second virtual view.

4. The method of claim 2, further comprising: for each of the plurality of cameras, determining (i) a first parallax confidence map associated with the first parallax map for the first virtual view based on the camera pose and the position of the camera in relation to the first virtual camera and (ii) a second parallax confidence map associated with the second parallax map for the second virtual view based on the camera pose and the position of the camera in relation to the second virtual camera; wherein the first parallax confidence map is used in generating the first virtual view and the second parallax confidence map is used in generating the second virtual view.

5. The method of claim 4, wherein: the first parallax confidence map is used to identify a priority of the first parallax map in generating the first virtual view; and the second parallax confidence map is used to identify a priority of the second parallax map in generating the second virtual view.

6. The method of claim 5, wherein a parallax map with a higher confidence than another parallax map is used in a front layer construction of a virtual view in relation to the other parallax map.

7. The method of claim 1, wherein: a confidence for the first or second confidence map is determined based on distance and orientation differences between the camera pose and the first or second virtual camera; and the confidence increases as the distance and orientation differences between the camera and the first or second virtual camera decrease.

8. A video see-through (VST) augmented reality (AR) device comprising: a plurality of cameras; and at least one processing device operably coupled to the cameras, the at least one processing device configured to: for each of the plurality of cameras: capture an image using the camera and associate the captured image with a camera pose; determine (i) a first contribution of the captured image for a first virtual view for display on a first display of the AR device and (ii) a second contribution of the captured image for a second virtual view for display on a second display of the AR device; and determine (i) a first confidence map for the first virtual view based on the camera pose and a position of the camera in relation to a first virtual camera and (ii) a second confidence map for the second virtual view based on the camera pose and the position of the camera in relation to a second virtual camera; and generate (i) the first virtual view by combining the first contribution using the first confidence map for each of the plurality of cameras and (ii) the second virtual view by combining the second contribution using the second confidence map for each of the plurality of cameras.

9. The VST AR device of claim 8, wherein: the at least one processing device is further configured, for each of the plurality of cameras, to determine a first parallax map for the first virtual camera and a second parallax map for the second virtual camera by projecting a depth map associated with the captured image onto image planes of the first and second displays; and the first parallax map is used in generating the first virtual view and the second parallax map is used in generating the second virtual view.

10. The VST AR device of claim 9, wherein: the first parallax map is used to identify a location of the first contribution in the first virtual view; and the second parallax map is used to identify a location of the second contribution in the second virtual view.

11. The VST AR device of claim 9, wherein: the at least one processing device is further configured, for each of the plurality of cameras, to determine (i) a first parallax confidence map associated with the first parallax map for the first virtual view based on the camera pose and the position of the camera in relation to the first virtual camera and (ii) a second parallax confidence map associated with the second parallax map for the second virtual view based on the camera pose and the position of the camera in relation to the second virtual camera; and the first parallax confidence map is used in generating the first virtual view and the second parallax confidence map is used in generating the second virtual view.

12. The VST AR device of claim 11, wherein: the first parallax confidence map is used to identify a priority of the first parallax map in generating the first virtual view; and the second parallax

confidence map is used to identify a priority of the second parallax map in generating the second virtual view.

13. The VST AR device of claim 12, wherein a parallax map with a higher confidence than another parallax map is used in a front layer construction of a virtual view in relation to the other parallax map.

14. The VST AR device of claim 8, wherein: a confidence for the first or second confidence map is determined based on distance and orientation differences between the camera pose and the first or second virtual camera; and the confidence increases as the distance and orientation differences between the camera and the first or second virtual camera decrease.

15. A non-transitory machine readable medium containing instructions that when executed cause at least one processor to: for each of a plurality of cameras of a video see-through (VST) augmented reality (AR) device: capture an image and associate the captured image with a camera pose; determine (i) a first contribution of the captured image for a first virtual view for display on a first display of the AR device and (ii) a second contribution of the captured image for a second virtual view for display on a second display of the AR device; and determine (i) a first confidence map for the first virtual view based on the camera pose and a position of the camera in relation to a first virtual camera and (ii) a second confidence map for the second virtual view based on the camera pose and the position of the camera in relation to a second virtual camera; and generate (i) the first virtual view by combining the first contribution using the first confidence map for each of the plurality of cameras and (ii) the second virtual view by combining the second contribution using the second confidence map for each of the plurality of cameras.

16. The non-transitory machine readable medium of claim 15, further containing instructions that when executed cause the at least one processor, for each of the plurality of cameras, to determine a first parallax map for the first virtual camera and a second parallax map for the second virtual camera by projecting a depth map associated with the captured image onto image planes of the first and second displays; wherein the first parallax map is used in generating the first virtual view and the second parallax map is used in generating the second virtual view.

17. The non-transitory machine readable medium of claim 16, wherein: the first parallax map is used to identify a location of the first contribution in the first virtual view; and the second parallax map is used to identify a location of the second contribution in the second virtual view.

18. The non-transitory machine readable medium of claim 16, further containing instructions that when executed cause the at least one processor, for each of the plurality of cameras, to determine (i) a first parallax confidence map associated with the first parallax map for the first virtual view based on the camera pose and the position of the camera in relation to the first virtual camera and (ii) a second parallax confidence map associated with the second parallax map for the second virtual view based on the camera pose and the position of the camera in relation to the second virtual camera; and wherein the first parallax confidence map is used in generating the first virtual view and the second parallax confidence map is used in generating the second virtual view.

19. The non-transitory machine readable medium of claim 18, wherein: the first parallax confidence map is used to identify a priority of the first parallax map in generating the first virtual view; the second parallax confidence map is used to identify a priority of the second parallax map in generating the second virtual view; and a parallax map with a higher confidence than another parallax map is used in a front layer construction of a virtual view in relation to the other parallax map.

20. The non-transitory machine readable medium of claim 15, wherein: a confidence for the first or second confidence map is determined based on distance and orientation differences between the camera pose and the first or second virtual camera; and the confidence increases as the distance and orientation differences between the camera and the first or second virtual camera decrease.

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